

The Impact of Generative Artificial Intelligence on Academic Performance, Critical Thinking, and Conceptual Understanding among High School Students in Dhaka: A Mixed-Methods Study

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Abstract

Bangladesh's secondary education system is marked by a high teacher-to-student ratio and faces many resource constraints; however, in the past few years, it has seen students increasingly turn to generative artificial intelligence (GenAI) tools such as ChatGPT, Gemini, and Claude for academic tasks. Yet, limited research exists on how reliance on GenAI tools actually affects cognitive and academic outcomes, especially in South Asia. This mixed-methods study has explored the relationship between GenAI reliance and four student outcomes: academic performance, critical thinking, creativity, and conceptual understanding.

The study was carried out among high school students aged 15–20 in Dhaka. Quantitative data were collected via a Google Forms survey ($N = 145$) which measured the frequency of GenAI use along with composite outcome scores and analyzed using Pearson and Spearman correlations. Qualitative data were gathered through semi-structured, asynchronous voice-message interviews with 12 purposively sampled students (one male and one female per age group, 15–20) and coded thematically in NVivo.

The frequency of using GenAI for academic purposes showed a significant positive correlation with self-reported academic performance ($r = .50, p < .001$) and conceptual understanding ($r = .44, p < .001$), while no significant association emerged with critical thinking ($r = -.10, p = .24$) or creativity ($r = .14, p = .09$). Qualitative findings complemented these results because the risk of overreliance and self-directed verification strategies were the two most frequently coded patterns (9/12 participants each). This suggested that students use both cognitive offloading and self-regulation at the same time, rather than splitting into two different groups. A majority of participants also framed GenAI as narrowing, rather than widening, educational access gaps (9/12 vs. 6/12). For this particular sample, the findings suggest that